

FARIHA SHAH

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Seattle, WA

Experience

Graduate Data Assistant

Oct 2025 – Present

Division of Mission Integration, Seattle University, Seattle, WA

- Designed and automated **5+** event data workflows using **Power Automate**, reducing manual coordination time by **60%** and compressing stakeholder response cycles from daily to hourly.
- Built and owned a **SharePoint** event database tracking **10K+** rows; integrated with **Tableau** to deliver interactive leadership dashboards, improving data accuracy by **30%**.

Data Engineer II

Jul 2020 – Aug 2024

FlowHCM, Karachi, Pakistan

- Diagnosed root cause of a critical HR data pipeline failure spanning **6 clients**; redesigned the ingestion layer from scratch, restoring executive reporting within **24 hours** and eliminating **10+ hours/week** of manual reconciliation.
- Architected a scalable HR KPI data platform integrating **50+** heterogeneous sources processing **30M+ rows/month** using **Python**, **PostgreSQL**, and **C#**; reduced analyst manual effort by **50%** and adopted as the org-wide reporting standard.
- Re-engineered **12** ETL pipelines on **SQL Server** processing **50M+** HR records using **C# .NET** incremental load patterns; cut pipeline runtimes by **40%** and infrastructure costs by **25%**, with zero data loss across rollout.

Data Analyst Intern

Jan 2020 – Jun 2020

Pakistan Petroleum Limited, Karachi, Pakistan

- Built **5** end-to-end ETL pipelines on large-scale geological datasets achieving **98%** data integrity; applied time-series forecasting using **Python** and **AWS SageMaker**, cutting reporting time by **15 hrs/week** and improving operational efficiency by **20%**.

Projects

Costco Wholesale Competitive Pricing Engine

Capstone

- Built a serverless data pipeline on **AWS** (Lambda, S3, EventBridge) ingesting daily competitor prices; developed a cross-price elasticity model across eggs, olive oil, and paper towels at **18 WA** locations paper towels showed highest sensitivity (elasticity **5.52**), enabling pricing recommendations with up to **31% projected volume lift**.

Automated Data Pipeline with AI-Driven Anomaly Explanation

- Designed an LLM-powered data quality monitoring pipeline on **AWS** (Lambda, S3) processing **100K+ records**; integrated **Claude API** to auto-generate natural-language anomaly explanations for analysts, reducing manual debugging time by **60%**.

Unstructured Data Ingestion & Semantic Search Engine

- Architected an **AWS** ingestion pipeline (Lambda, S3, **Bedrock** Titan Embeddings) processing **500+ daily documents** into **10K+** pgvector vectors on **RDS**; exposed a **FastAPI** semantic search endpoint, replacing manual document retrieval entirely.

Retail Price Forecasting

- Built a large-scale AWS data pipeline scraping **3.8M+** daily grocery prices from Walmart and Kroger across **57 WA** locations; engineered **14-day** price forecasts using **Facebook Prophet** (**MAPE < 4%**) and delivered results on a live **Streamlit** dashboard covering the full data engineering lifecycle.

Education

M.S. in Data Science

June 2026

Seattle University, Seattle, WA

B.E. in Petroleum & Natural Gas Engineering

2016 – 2019

Dawood University of Engineering & Technology, Pakistan

Skills

Languages: Python, SQL, R, C#, .NET

Cloud & Data Infra: Lambda, S3, EMR, Redshift, Glue, Athena, DynamoDB, EventBridge, SageMaker, Bedrock, QuickSight, Docker, Snowflake

Data & ETL: ETL, Data Modeling, Data Warehousing, REST APIs, DBT, Kafka, PySpark, Apache Spark, Hadoop, FastAPI

ML & Analytics: scikit-learn, TensorFlow, PyTorch, Time-Series Forecasting, A/B Testing, Predictive Modeling

BI & Workflow: Tableau, Power BI, Power Automate, Excel

Dev Tools: Git, Bash, UNIX, GitHub Actions, Claude API